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1 Boundary Conditions

1.1 Moment based boundary conditions

Moment based boundary conditions in 3 dimensions are investigated in [3], [2]. The idea is to approximate raw moments as their Maxwell-Boltzman equilibrium and use those in a system of equations to calculate the unknown distributions. The equilibriums are calculated from ρ, u_x, u_y and u_z . It is required that those 4 quantities are known a priori by prescribing them or by restoring them from known distributions. A general method to restore macroscopic quantities from known distributions is shown later on.

One advantage of using moment based boundary conditions is locality [3]. It is only necessary to work with information from a single cell.

Another advantage is that the prescribed quantities have clearer physical meaning compared to some other options, where part of the definition is somewhat artificial [3].

Derivation of moment based boundary conditions is explained below.

1.1.1 General procedure

Raw moments are defined as [1]

$$m_{abc} = \sum_i f_i c_{xi}^a c_{yi}^b c_{zi}^c = \sum_{\text{known } i} f_i c_{xi}^a c_{yi}^b c_{zi}^c + \sum_{\text{unknown } i} f_i c_{xi}^a c_{yi}^b c_{zi}^c \quad (1.1)$$

Each raw moment is just a linear combination of distributions f , from which some are known and some are unknown.

Equilibrium moments can be expressed as

$$m_{abc}^{eq} = \sum_i f_i^{eq} c_{xi}^a c_{yi}^b c_{zi}^c \quad (1.2)$$

where f_i^{eq} is the Maxwell-Boltzman equilibrium:

$$f_i^{eq} = w_i \rho \left[1 + 3 \underline{c}_i \cdot \underline{u} + \frac{9}{2} (\underline{c}_i \cdot \underline{u})^2 - \frac{3}{2} \underline{u} \cdot \underline{u} \right]. \quad (1.3)$$

Maxwell-Boltzman equilibrium moments relevant to the task are shown in Table 1. For use in a later chapter, there is also a linearization shown where any members with second or higher power of u are neglected, which is reasonable if $Ma \ll 1$.

Table 1: Maxwell–Boltzmann equilibrium moments and linearized equilibrium

Moment	Maxwell–Boltzmann equilibrium	Linearized equilibrium
m_{000}	ρ	ρ
m_{100}	ρu_x	ρu_x
m_{010}	ρu_y	ρu_y
m_{001}	ρu_z	ρu_z
m_{200}	$\frac{1}{3}\rho + \rho u_x^2$	$\frac{1}{3}\rho$
m_{020}	$\frac{1}{3}\rho + \rho u_y^2$	$\frac{1}{3}\rho$
m_{002}	$\frac{1}{3}\rho + \rho u_z^2$	$\frac{1}{3}\rho$
m_{011}	$\rho u_y u_z$	0
m_{101}	$\rho u_x u_z$	0
m_{110}	$\rho u_x u_y$	0
m_{111}	0	0
m_{210}	$\frac{1}{3}\rho u_y + \rho u_x^2 u_y$	$\frac{1}{3}\rho u_y$
m_{201}	$\frac{1}{3}\rho u_z + \rho u_x^2 u_z$	$\frac{1}{3}\rho u_z$
m_{120}	$\frac{1}{3}\rho u_x + \rho u_x u_y^2$	$\frac{1}{3}\rho u_x$
m_{021}	$\frac{1}{3}\rho u_z + \rho u_y^2 u_z$	$\frac{1}{3}\rho u_z$
m_{102}	$\frac{1}{3}\rho u_x + \rho u_x u_z^2$	$\frac{1}{3}\rho u_x$
m_{012}	$\frac{1}{3}\rho u_y + \rho u_y u_z^2$	$\frac{1}{3}\rho u_y$
m_{022}	$\frac{1}{9}\rho + \frac{1}{3}\rho u_y^2 + \frac{1}{3}\rho u_z^2 + \rho u_y^2 u_z^2$	$\frac{1}{9}\rho$
m_{202}	$\frac{1}{9}\rho + \frac{1}{3}\rho u_x^2 + \frac{1}{3}\rho u_z^2 + \rho u_x^2 u_z^2$	$\frac{1}{9}\rho$
m_{220}	$\frac{1}{9}\rho + \frac{1}{3}\rho u_x^2 + \frac{1}{3}\rho u_y^2 + \rho u_x^2 u_y^2$	$\frac{1}{9}\rho$
m_{211}	$\frac{1}{3}\rho u_y u_z$	0
m_{121}	$\frac{1}{3}\rho u_x u_z$	0
m_{112}	$\frac{1}{3}\rho u_x u_y$	0

It is required that ρ, u_x, u_y, u_z are defined a priori, then all equilibrium moments can be calculated.

The idea is to set raw moments from equation (1.1) equal to their Maxwell-Boltzmann equilibrium from Table 1. This gives a system of linear equations that allows to find unknown distributions:

$$\sum_{\text{known } i} f_i c_{xi}^a c_{yi}^b c_{zi}^c + \sum_{\text{unknown } i} f_i c_{xi}^a c_{yi}^b c_{zi}^c = m_{abc}^{eq} \quad (1.4)$$

Approximating the moments as their equilibrium is exact for raw moments up to first order. For higher order moments, using the equilibrium means that the non-equilibrium part is neglected. This approximation still recovers incompressible Navier-Stokes equations correctly [2].

The specific formulation of equations depends on type and orientation of the cell. A face cell is a cell that does not have neighbours in one direction. A edge cell does not have neighbours in two directions. A corner cell misses neighbours in three directions. A general approach to the derivation process is needed to handle all different kinds of cells [2]. A section showing the equations for one of the cell orientations as an example, follows. The procedure presented below can be algorithmized and used to automatically generate code for all different cell types.

1.1.2 Corner cell example

As an example to explain the procedure, let's use the most "difficult" case of a corner cell with outer normal $(1, -1, 1)$ (so it has neighbours in X direction but misses neighbours in positive Y and negative Z). After streaming, the distributions with $c_x = -1, c_y = 1$ or $c_z = -1$ are unknown.

Let's put all known distributions into a vector $\underline{f_k}$ and unknown distributions into $\underline{f_u}$ as displayed in Table 2.

Table 2: Known and unknown distributions for a cell with $(1, -1, 1)$ outer normal

Known distributions	$\underline{f_k} = (f_0, f_1, f_4, f_5, f_9, f_{14}, f_{16}, f_{20})^T$
Unknown distributions	$\underline{f_u} = (f_2, f_3, f_6, f_7, f_8, f_{10}, f_{11}, f_{12}, f_{13}, f_{15}, f_{17}, f_{18}, f_{19}, f_{21}, f_{22}, f_{23}, f_{24}, f_{25}, f_{26})^T$

Equilibrium moments will be stored in a vector $\underline{m} \in \mathbb{R}^{23}$:

Table 3: Vector of equilibrium moments

$$\text{Equilibrium moments} \quad \underline{m} = \begin{pmatrix} m_{000}, m_{100}, m_{010}, m_{001}, \\ m_{200}, m_{020}, m_{002}, m_{011}, m_{101}, m_{110}, \\ m_{111}, m_{210}, m_{201}, m_{120}, m_{021}, m_{102}, m_{012}, \\ m_{022}, m_{202}, m_{220}, m_{211}, m_{121}, m_{112} \end{pmatrix}^T$$

This allows to express equation 1.4 in a matrix form:

$$\underline{\underline{M_{fk}}} \underline{f_k} + \underline{\underline{M_{fu}}} \underline{f_u} = \underline{m} \quad (1.5)$$

where $\underline{\underline{M_{fk}}} \in \mathbb{Z}^{23 \times \text{len}(\underline{f_k})}$ is the matrix of f_k coefficients:

$$\underline{\underline{M_{fk}}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 & 0 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \quad (1.6)$$

and $\underline{\underline{M}}_{fu} \in \mathbb{Z}^{23 \times \text{len}(f_u)}$ is the matrix of f_u coefficients:

$$\underline{\underline{M}}_{fu} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 0 & 1 & -1 & -1 & -1 & 1 & 0 & -1 & 0 & 0 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 1 & 1 & 1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ 0 & -1 & 0 & -1 & 1 & -1 & 0 & 0 & -1 & 0 & 1 & -1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & -1 & 0 & 0 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & -1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & -1 & 0 & 0 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & -1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 \end{bmatrix} \quad (1.7)$$

Equation (1.5) gives 23 scalar equations. For a corner cell with outer normal $(1, -1, 1)$ there are 19 unknown distributions, so from (1.5) it is necessary to pick 19 equations that have linearly independent rows of $\underline{\underline{M}}_{fu}$. It is suggested to select those corresponding to lower order moments, as they have clearer physical meaning [3].

To select the equations, let's define a selector matrix $\underline{\underline{S}} \in \mathbb{Z}^{\text{len}(f_u) \times 23}$. $\underline{\underline{S}}$ looks almost like an identity matrix, but 4 rows are left out to only keep the desired 19 equations:

$$\underline{\underline{S}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (1.8)$$

By multiplying (1.5) by $\underline{\underline{S}}$, the correct 19 equations are selected:

$$\underline{\underline{S}}\underline{\underline{M}}_{fk}f_k + \underline{\underline{S}}\underline{\underline{M}}_{fu}f_u = \underline{\underline{S}}m \quad (1.9)$$

$\left(\underline{\underline{SM_{fu}}}\right)$ is regular, so it can be inverted. This allows to find $\underline{f_u}$:

$$\underline{f_u} = \left(\underline{\underline{SM_{fu}}}\right)^{-1} \left(\underline{\underline{Sm}} - \underline{\underline{SM_{fk}f_k}}\right) \quad (1.10)$$

As $\left(\underline{\underline{S}} \ \underline{\underline{M_{fu}}}\right)$ is relatively small and nice, it can be inversed using a symbolic solver.

$$\left(\underline{\underline{S}} \ \underline{\underline{M_{fu}}}\right)^{-1} =$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & -1/2 & 0 & 0 & 0 & 0 & 0 & -1/2 \\ 0 & 0 & 0 & 1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & -1/2 & 0 & 0 & -1/2 & 0 & 0 \\ -1 & -1/2 & 0 & -1/2 & 1/2 & 1 & 1/2 & 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 & -1/2 & -1/2 \\ 0 & 0 & -1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & -1/2 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/2 & 0 & 0 & 0 & -1/2 \\ -1 & 0 & 1/2 & -1/2 & 1 & 1/2 & 1/2 & 0 & 0 & 0 & -1/2 & 1/2 & 0 & 0 & 0 & 0 & -1/2 & -1/2 \\ -1 & -1/2 & 1/2 & 0 & 1/2 & 1/2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/2 & -1/2 & -1/2 & -1/2 \\ 0 & 0 & 0 & 1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & -1/2 & 0 & 0 & 0 & 0 & 0 & 0 & -1/2 \\ 0 & 0 & -1/2 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 & -1/2 \\ 1 & 1/2 & -1/2 & 1/2 & -1/2 & -1/2 & -1/2 & -1/4 & 1/4 & -1/4 & 1/4 & -1/4 & -1/4 & -1/4 & -1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/4 & 0 & 0 & 0 & 0 & 0 & 1/4 & 0 & -1/4 & 1/4 & 0 \\ 0 & -1/2 & 0 & 0 & -1/2 & 0 & 0 & 0 & -1/4 & 1/4 & 1/4 & -1/4 & 1/4 & 0 & 1/4 & 0 & 0 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1/4 & -1/4 & 0 & 1/4 & 0 & 0 & 0 & 0 & 1/4 \\ 0 & 0 & 0 & -1/2 & 0 & 0 & -1/2 & 1/4 & -1/4 & 0 & 0 & 1/4 & 0 & 1/4 & -1/4 & 1/4 & 1/4 & 0 \\ 0 & 0 & 1/2 & 0 & 0 & -1/2 & 0 & 1/4 & 0 & 1/4 & -1/4 & 0 & -1/4 & -1/4 & 0 & -1/4 & 1/4 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1/4 & 0 & 0 & 1/4 & 0 & 0 & 1/4 & 0 & 0 & 1/4 \end{bmatrix} \quad (1.11)$$

1.2 Reconstruction of macroscopic quantities from known distributions

The moment based boundary conditions allow to calculate $\underline{f_u}$ from $\underline{f_k}$ and macroscopic quantities ρ, u_x, u_y, u_z . In case that not all macroscopic quantities are explicitly prescribed, it is possible to reconstruct them from $\underline{f_k}$.

1.2.1 Corner cell example with unknown ρ, u_x, u_y, u_z

Let's consider a corner cell with $(1, -1, 1)$ outer normal again. Let's first illustrate a case where no macroscopic quantity is prescribed, to show how it is possible to reconstruct all ρ, u_x, u_y, u_z just from $\underline{f_k}$.

Known and unknown distributions for such case were previously displayed in Table 2. Matrices $\underline{\underline{M_{fk}}}$ and $\underline{\underline{M_{fu}}}$ were listed in equations (1.6) and (1.7), respectively.

In the previous chapter, the goal was to calculate $\underline{f_u}$, which required finding linearly independeng rows of $\underline{\underline{M_{fu}}}$. Now the situation is reversed: goal is to find at least 4 independent equations for ρ, u_x, u_y, u_z where no unknown $\underline{f_i}$ is present.

This can be achieved by forming linear combinations of the available equations. Let $\underline{c} \in \mathbb{R}^{23}$ denote a column vector of coefficients, where each component multiplies the corresponding equation. Then the unknown distributions get eliminated if for all $\underline{f_u}$ it holds that

$$\left(\underline{\underline{M_{fu}}} \ \underline{f_u}\right) \cdot \underline{c} = 0 \quad (1.12)$$

This can be reformulated as

$$\left(\underline{\underline{M_{fu}}} \ \underline{f_u}\right) \cdot \underline{c} = \underline{c}^T \left(\underline{\underline{M_{fu}}} \ \underline{f_u}\right) = \left(\underline{c}^T \ \underline{\underline{M_{fu}}}\right) \underline{f_u} = \underline{f_u}^T \left(\underline{c}^T \ \underline{\underline{M_{fu}}}\right)^T = \underline{f_u}^T \left(\underline{\underline{M_{fu}}}^T \ \underline{c}\right) = 0 \quad (1.13)$$

which will hold for all $\underline{f_u}$ only if

$$\underline{\underline{M_{fu}}}^T \ \underline{c} = \underline{0} \quad (1.14)$$

Therefore, the searched coefficient vectors $\underline{c}$ are given by the null space of $\underline{\underline{M}}_{fu}^T$. Let's assemble all n linearly independent vectors $\underline{c}$ into a matrix $\underline{\underline{C}}$:

$$\underline{\underline{C}} = \begin{bmatrix} | & | & | & | \\ \underline{c}_0 & \underline{c}_1 & \dots & \underline{c}_n \\ | & | & | & | \end{bmatrix} \quad (1.15)$$

where $\underline{c}_0, \underline{c}_1 \dots$ are ordered smallest to largest using the key

$$\text{key} = \sum_i^{\text{len}(\underline{c})} w_i, \quad w_i = \begin{cases} \text{order of moment } m_i & \text{if } c_i \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (1.16)$$

which puts low order moment combinations first. In case of outer normal $(1, -1, 1)$, $\underline{\underline{C}}$ yields

$$\underline{\underline{C}} = \begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & 0 & -1 & 1 \\ 1 & -1 & 0 & -1 \\ -1 & 1 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.17)$$

That suggests how to deal with f_u . To relate remaining f_k to ρ, u_x, u_y, u_z , equilibrium moments will be used once again. However, Maxwell-Boltzman equilibrium moments as shown in Table 1 are nonlinear. Therefore the linearized equilibrium will be used to provide a linear set of equations.

Let's define a vector of independent macroscopic quantities $\underline{q}$:

$$\underline{q} = (\rho, \rho u_x, \rho u_y, \rho u_z)^T \quad (1.18)$$

Then linearized equilibrium moments can be expressed as

$$\underline{m} = \underline{\underline{Q}} \underline{q} \quad (1.19)$$

where $\underline{\underline{Q}} \in \mathbb{Z}^{23 \times 4}$.

$$\underline{\underline{Q}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ \frac{1}{9} & 0 & 0 & 0 \\ \frac{1}{9} & 0 & 0 & 0 \\ \frac{1}{9} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (1.20)$$

Now, inserting (1.19) into (1.5) and multiplying both sides from the left by $\underline{\underline{C}}^T$ gives

$$\underline{\underline{C}}^T \left(\underline{\underline{M}}_{fk} \underline{f}_k + \underline{\underline{M}}_{fu} \underline{f}_u \right) = \underline{\underline{C}}^T \underline{\underline{Q}} \underline{q} \quad (1.21)$$

As follows from above, $\underline{\underline{C}}^T$ is built so that $\underline{f}_u$ gets cancelled, so

$$\underline{\underline{C}}^T \underline{\underline{M}}_{fk} \underline{f}_k = \underline{\underline{C}}^T \underline{\underline{Q}} \underline{q} \quad (1.22)$$

Equation (1.22) provides n scalar equations for n vectors $\underline{c}$. The unknown is $\underline{q}$. In a general case it may happen that n is higher than 4 which is the number of unknowns. Therefore we need to select such equations that have linearly independent rows of $\left(\underline{\underline{C}}^T \underline{\underline{Q}} \right)$ and which use the lowest order moments.

Thanks to the ordering (1.16), the preferred rows are placed on top.

To select the equations, let's use a new selector matrix $\underline{\underline{S}}_q \in \mathbb{Z}^{4 \times \text{columns}(\underline{\underline{C}})}$. In case of outer normal $(1, -1, 1)$, $\underline{\underline{S}}$ yields:

$$\underline{\underline{S}}_q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1.23)$$

Multiplying (1.22) by $\underline{\underline{S}}_q$ selects the correct equations:

$$\underline{\underline{S}}_q \underline{\underline{C}}^T \underline{\underline{M}}_{fk} \underline{f}_k = \underline{\underline{S}}_q \underline{\underline{C}}^T \underline{\underline{Q}} \underline{q} \quad (1.24)$$

where $\left(\underline{\underline{S}}_q \underline{\underline{C}}^T \underline{\underline{Q}} \right)$ is regular, so it can be inverted. This allows to find $\underline{q}$:

$$\underline{q} = \left(\underline{\underline{S}}_q \underline{\underline{C}}^T \underline{\underline{Q}} \right)^{-1} \underline{\underline{S}}_q \underline{\underline{C}}^T \underline{\underline{M}}_{fk} \underline{f}_k \quad (1.25)$$

As $\left(\underline{\underline{S_q C^T Q}}\right)$ is relatively small and nice, it can be inversed using a symbolic solver.

$$\left(\underline{\underline{S_q C^T Q}}\right)^{-1} = \begin{bmatrix} 2 & 3/2 & -3/2 & 3/2 \\ -1 & -3/2 & 0 & 0 \\ 1 & 0 & -3/2 & 0 \\ -1 & 0 & 0 & -3/2 \end{bmatrix} \quad (1.26)$$

1.2.2 Corner cell example with prescribed ρ or u_x, u_y, u_z

Let's modify the procedure for a case when either ρ is given or u_x, u_y, u_z are given. Compared to the previous derivation, $\underline{f_k}, \underline{f_u}, \underline{\underline{M_{fk}}}, \underline{\underline{M_{fu}}}, \underline{C}$ stay the same. However, the vector of independent macroscopic quantities should be split in the ρ and u part. Let's define

$$\underline{u} = (u_x, u_y, u_z)^T \quad (1.27)$$

which results in new expression of $\underline{m}$:

$$\underline{m} = \rho \left(\underline{\underline{Q_u}} \underline{u} + \underline{Q_\rho} \right) \quad (1.28)$$

where

$$\underline{\underline{Q_u}} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (1.29)$$

and

$$\underline{Q}_\rho = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{9} \\ \frac{1}{9} \\ \frac{1}{9} \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (1.30)$$

That leads to

$$\underline{C}^T \underline{M}_{fk} \underline{f}_k = \underline{C}^T \rho \left(\underline{Q}_u \underline{u} + \underline{Q}_\rho \right) \quad (1.31)$$

reorganized

$$\frac{1}{\rho} \underline{C}^T \underline{M}_{fk} \underline{f}_k - \underline{C}^T \underline{Q}_\rho = \underline{C}^T \underline{Q}_u \underline{u} \quad (1.32)$$

From here the case splits depending on if ρ or $\underline{u}$ is the unknown. If $\underline{u}$ is the unknown, we need to select such equations that have linearly independent rows of $\left(\underline{C}^T \underline{Q}_u \right)$. For that let's use a selector matrix $\underline{S}_u \in \mathbb{Z}^{4 \times \text{columns}(\underline{C})}$:

$$\underline{S}_u = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (1.33)$$

Then

$$\underline{u} = \left(\underline{S}_u \underline{C}^T \underline{Q}_u \right)^{-1} \left[\frac{1}{\rho} \underline{S}_u \underline{C}^T \underline{M}_{fk} \underline{f}_k - \underline{S}_u \underline{C}^T \underline{Q}_\rho \right] \quad (1.34)$$

where

$$\left(\underline{S}_u \underline{C}^T \underline{Q}_u \right)^{-1} = \begin{bmatrix} -1 & -3/2 & 0 \\ 1 & 0 & -3/2 \\ 1 & 3/2 & -3/2 \end{bmatrix} \quad (1.35)$$

If ρ is the unknown in (1.32), we can manipulate (1.32) to

$$\underline{C}^T \underline{M}_{fk} \underline{f}_k = \rho \left[\underline{C}^T \underline{Q}_\rho + \underline{C}^T \underline{Q}_u \underline{u} \right] \quad (1.36)$$

To isolate ρ , we need to define $\underline{S}_\rho \in \mathbb{Z}^{1 \times \text{columns}(\underline{C})}$ so that it selects a single equation:

$$\underline{S}_\rho \underline{C}^T \underline{M}_{fk} \underline{f}_k = \rho \left[\underline{S}_\rho \underline{C}^T \underline{Q}_\rho + \underline{S}_\rho \underline{C}^T \underline{Q}_u \underline{u} \right] \quad (1.37)$$

Here we need to ensure that $\underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_\rho + \underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_u \underline{u}$ is not 0. The dependency on $\underline{u}$ is uncomfortable, but by looking at size of elements in $\underline{Q}_\rho, \underline{Q}_u$ and assuming that $\text{Ma} \ll 1$, it should be sufficient to ensure that $\underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_\rho$ is not 0. This is satisfied for the present case by

$$\underline{\underline{S}}_\rho = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad (1.38)$$

Then

$$\rho = \left(\underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_\rho + \underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_u \underline{u} \right)^{-1} \underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{M}_{fk} \underline{f}_k \quad (1.39)$$

Note that $\left(\underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_\rho + \underline{\underline{S}}_\rho \underline{\underline{C}}^T \underline{Q}_u \underline{u} \right)$ is a scalar, so there is no matrix inversion needed.

1.3 Non-reflecting boundary conditions

The work by Schlaffer [4] derives the following relation for the left boundary:

$$\left(\rho^{(n)} - \rho^{(n-1)} \right) c_s^2 + \frac{1}{2} \rho^{(n)} \left(u_x^{(n)} - u_x^{(n-1)} \right)^2 + \left(u_x^{(n)} - u_x^{(n-1)} \right) \rho^{(n)} c_s = 0 \quad (1.40)$$

To derive a general rule that works for any axis-aligned boundary face, for both inlets and outlets, let's define the normal velocity as

$$u_n = \underline{u} \cdot \underline{n} \quad (1.41)$$

also define

$$c_{ni} = \underline{c}_i \cdot \underline{n} \quad (1.42)$$

where $\underline{n}$ is the outer normal pointing out of the domain. Then (1.40) becomes

$$\left(\rho^{(n)} - \rho^{(n-1)} \right) c_s^2 + \frac{1}{2} \rho^{(n)} \left(u_n^{(n)} - u_n^{(n-1)} \right)^2 - \left(u_n^{(n)} - u_n^{(n-1)} \right) \rho^{(n)} c_s = 0 \quad (1.43)$$

For a facial cell (not an edge cell or a corner cell), the known distributions, ρ and u_n are connected by

$$\rho = \sum_{c_{ni}=-1} f_i + \sum_{c_{ni}=0} f_i + \sum_{c_{ni}=1} f_i \quad (1.44)$$

$$\rho u_n = - \sum_{c_{ni}=-1} f_i + \sum_{c_{ni}=1} f_i \quad (1.45)$$

By eliminating the unknown sum with $c_{ni} = -1$, we get

$$\rho(1 + u_n) = \sum_{c_{ni}=0} f_i + 2 \sum_{c_{ni}=1} f_i \quad (1.46)$$

Following convention of [4], label

$$\rho_z = \sum_{c_{ni}=0} f_i + 2 \sum_{c_{ni}=1} f_i \quad (1.47)$$

and so

$$u_n = \frac{\rho_z}{\rho} - 1 \quad (1.48)$$

Plug this for $u_n^{(n)}$ in (1.43), then

$$\left(\rho^{(n)} - \rho^{(n-1)} \right) c_s^2 + \frac{1}{2} \rho^{(n)} \left(\left(\frac{\rho_z^{(n)}}{\rho^{(n)}} - 1 \right) - u_n^{(n-1)} \right)^2 - \left(\left(\frac{\rho_z^{(n)}}{\rho^{(n)}} - 1 \right) - u_n^{(n-1)} \right) \rho^{(n)} c_s = 0 \quad (1.49)$$

We need to solve $\rho^{(n)}$ which is the only unknown.

$$\begin{aligned} & \left(\rho^{(n)} \right)^2 \left[\frac{1}{2} \left(u_n^{(n-1)} \right)^2 + (c_s + 1) u_n^{(n-1)} + c_s^2 + c_s + \frac{1}{2} \right] \\ & + \left(\rho^{(n)} \right) \left[-\rho^{(n-1)} c_s^2 - \rho_z^{(n)} \left(u_n^{(n-1)} + c_s + 1 \right) \right] + \frac{1}{2} \left(\rho_z^{(n)} \right)^2 = 0 \end{aligned} \quad (1.50)$$

$$\rho_{1,2}^{(n)} = \frac{\rho^{(n-1)}c_s^2 + \rho_z^{(n)}\left(u_n^{(n-1)} + c_s + 1\right) \pm c_s \sqrt{\left(\rho^{(n-1)}\right)^2 c_s^2 + 2\rho^{(n-1)}\rho_z^{(n)}\left(u_n^{(n-1)} + c_s + 1\right) - \left(\rho_z^{(n)}\right)^2}}{\left(u_n^{(n-1)} + c_s + 1\right)^2 + c_s^2} \quad (1.51)$$

To choose the correct branch, we require that if $u_n^{(n)} = u_n^{(n-1)}$, then also $\rho^{(n)} = \rho^{(n-1)}$. We plug this into (1.48), which gives

$$\rho_z^{(n)} = \rho^{(n-1)}\left(u_n^{(n-1)} + 1\right) \quad (1.52)$$

Inserting this into (1.51) and simplifying,

$$\rho_{1,2}^{(n)} = \frac{\rho^{(n-1)}c_s^2 + \rho^{(n-1)}\left(u_n^{(n-1)} + 1\right)\left(u_n^{(n-1)} + c_s + 1\right) \pm c_s \sqrt{\left(\rho^{(n-1)}\right)^2 \left(u_n^{(n-1)} + c_s + 1\right)^2}}{\left(u_n^{(n-1)} + c_s + 1\right)^2 + c_s^2} \quad (1.53)$$

Because under the square root $\rho^{(n-1)} > 0$ and $\left(u_n^{(n-1)} + c_s + 1\right) > 0$,

$$\rho_{1,2}^{(n)} = \rho^{(n-1)} \frac{c_s^2 + \left(u_n^{(n-1)} + c_s + 1\right)\left(u_n^{(n-1)} + 1\right) \pm c_s \left(u_n^{(n-1)} + c_s + 1\right)}{c_s^2 + \left(u_n^{(n-1)} + c_s + 1\right)^2} \quad (1.54)$$

For the positive branch, numerator equals the denominator and satisfies $\rho^{(n)} = \rho^{(n-1)}$.

2 References

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